

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- This week: Change of basis; finishing determinants; starting Chapter 4.
- Requizzing today 3:30-4:30 in Eckhart 207A, Quizzes 1-5
- HW₃ debrief is Wednesday 4-5pm, Eckhart 207A
- Quizzes 6 & 7 on Friday
- HW₄ due Friday / Saturday.
- Midterm exam review next Sunday
- Midterm exam next Monday

Today:

three new notions of equivalence for matrices (linear transformations):

column equivalence, matrix equivalence, and similarity

process to change bases in a vector space

a fundamental issue with coordinates: alias vs. alibi

I'm taking linear combinations of your columns!

$$\begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 \\ 0 & 8 & 0 \\ 2 & 4 & 1 \end{pmatrix}$$

No, I'm taking linear combinations of your rows!

$$\begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 \\ 0 & 8 & 0 \\ 2 & 4 & 1 \end{pmatrix}$$

I'm taking linear combinations of your columns!

I'm taking linear combinations of its rows!

I'm taking linear combinations of its columns!

$$\begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 \\ 0 & 8 & 0 \\ 2 & 4 & 1 \end{pmatrix}$$

row equivalence

Two matrices are equivalent
if they are connected
by a sequence of row operations.

$$A = E_3 E_2 E_1 B$$

matrix equivalence

Two matrices are equivalent
if they are connected
by a sequence of row operations
and/or column operations.

$$A = E_3 E_2 E_1 B E_4 E_5 E_6$$

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$$A = B E_4 E_5 E_6$$

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changes basis in codomain

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changes basis in domain
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$$A = E_3 E_2 E_1 B$$

changes basis in codomain

$$A = QB$$

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changes basis in domain
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$$A = QBP$$

column equivalence

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$$A = B E_4 E_5 E_6$$

changes basis in domain

$$A = BP$$

Row reduction of a matrix A ...

PRESERVES:

its nullspace (= solution set)

its nullity (= $\dim(\text{Null } A)$)

its row space

its rank (= $\dim(\text{Col } A) = \dim(\text{range})$)

whether or not columns span (or: are LI)

invertibility

CAN CHANGE:

the matrix (and corresponding LT)

its column space (= range)

the inverse

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PRESERVES:

nullspace (= solution set)

row space

nullity and rank

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PRESERVES:

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PRESERVES:

left null space (= $\text{Ker } A^T$)
(4th fundamental subspace)

column space (= range)

nullity and rank

The geometry of matrix equivalence

$$C : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

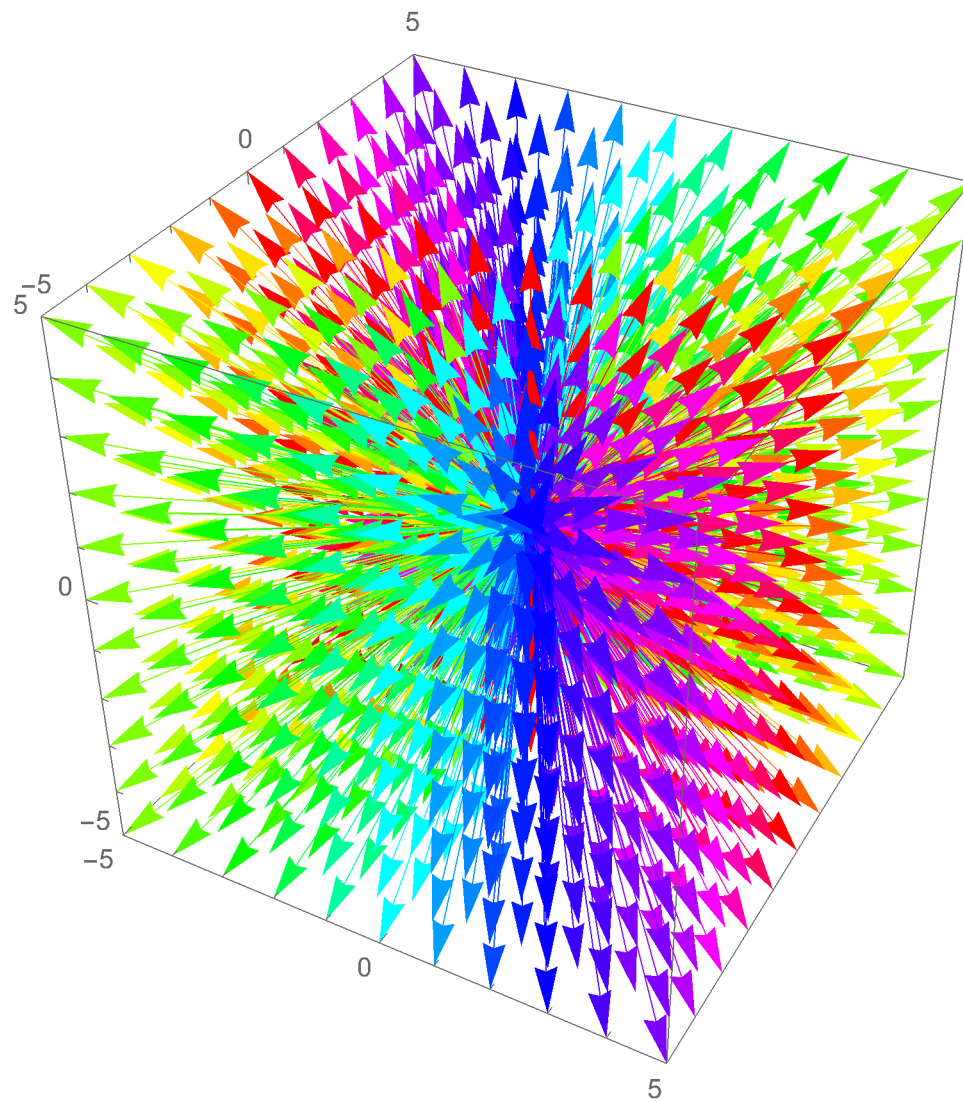
(column operation, invertible)

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

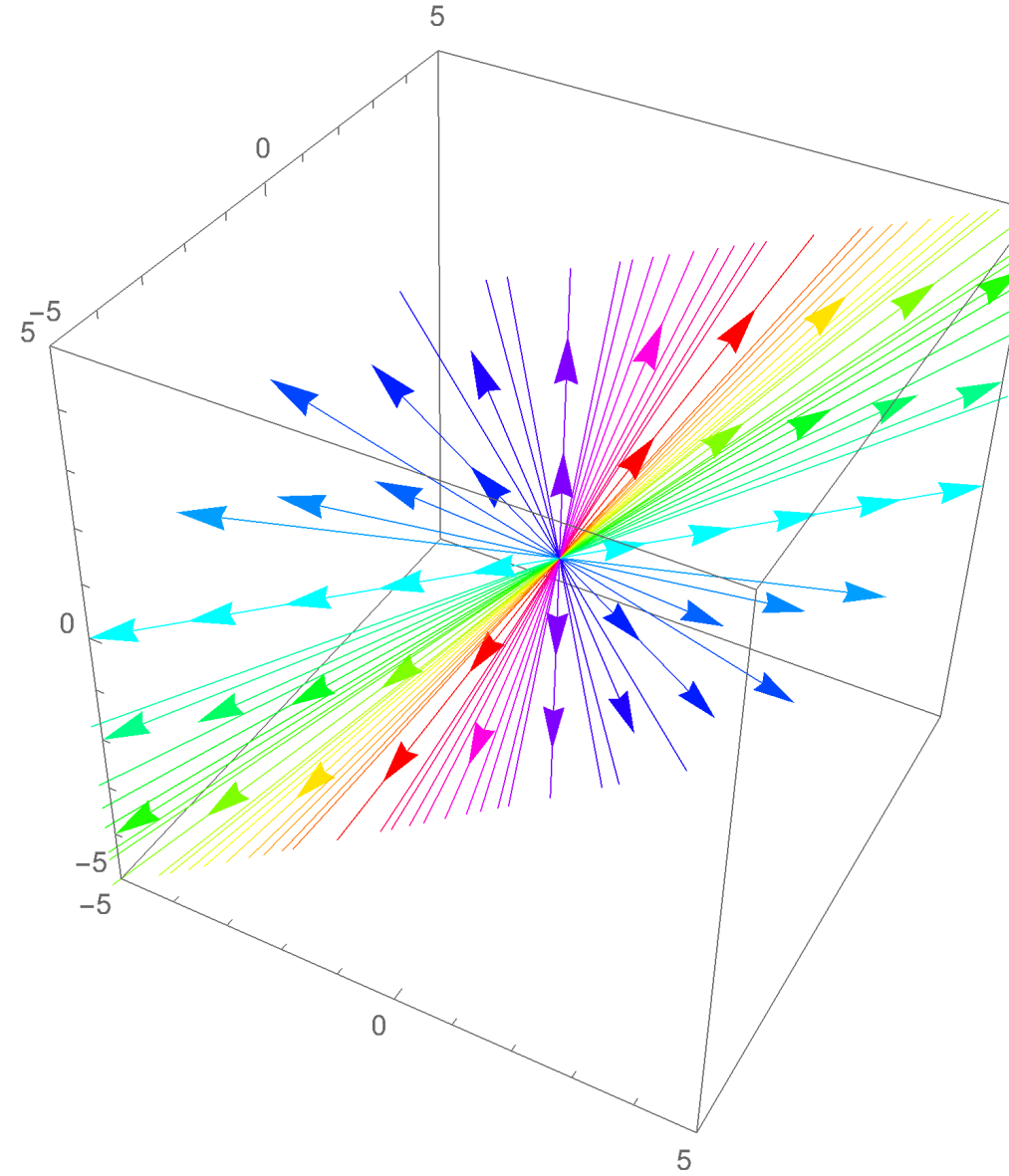
$$R : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

(row operation, invertible)

C ↻



A →



↻ R

RAC

A vs. RAC : nullspace and range can change,
but nullity and rank stay the same

Example of reducing to a simple matrix equivalence form:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 12 \\ 3 & 9 & 12 \end{pmatrix}$$

Example of reducing to a simple matrix equivalence form:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 12 \\ 3 & 9 & 12 \end{pmatrix} \xleftrightarrow{\text{RR}} \begin{pmatrix} 1 & 0 & -8 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{pmatrix}$$

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Theorem: Every matrix is matrix equivalent to a unique matrix with 1's for some way along its main diagonal, and 0's everywhere else. $A = P "I" Q$

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Theorem: Every matrix is matrix equivalent to a unique matrix with 1's for some way along its main diagonal, and 0's everywhere else. $A = P "I" Q$

Pf: Row reduce to RREF. Then column swap to get all k pivots in the first k columns. Then use pivots to eliminate all non-zero entries that are not pivots.

row equivalence

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & * \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

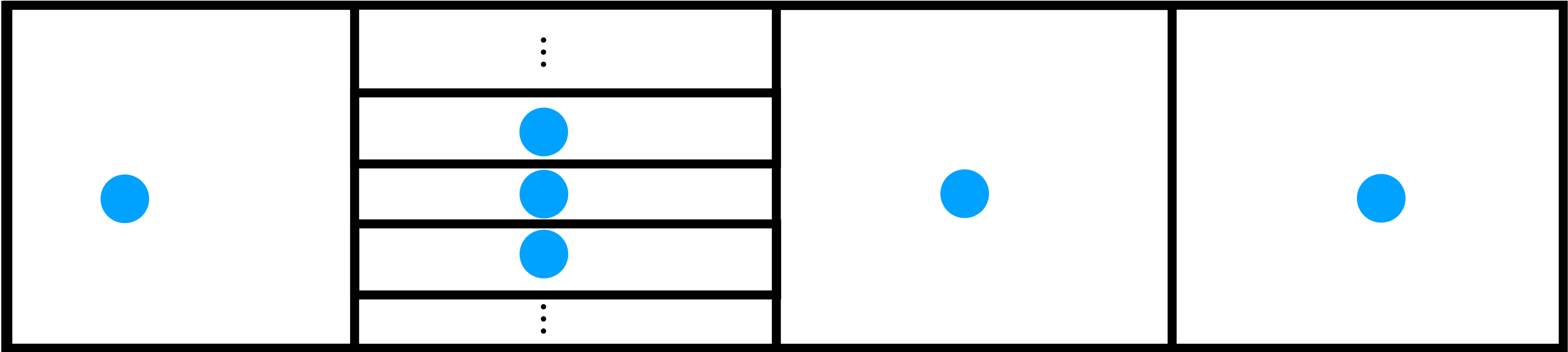
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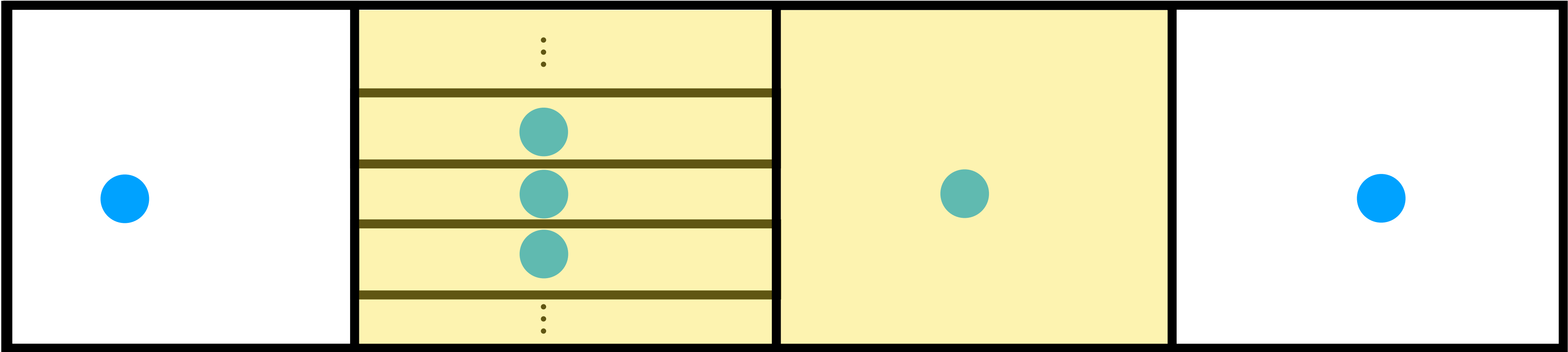
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$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

matrix equivalence

Matrix equivalence lets us choose our basis in the domain and codomain independently.

For an operator (linear transformation from V to itself), it makes sense to choose an arbitrary basis, but to keep it the same for the domain and codomain

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For an operator (linear transformation from V to itself), it makes sense to choose an arbitrary basis, but to keep it the same for the domain and codomain

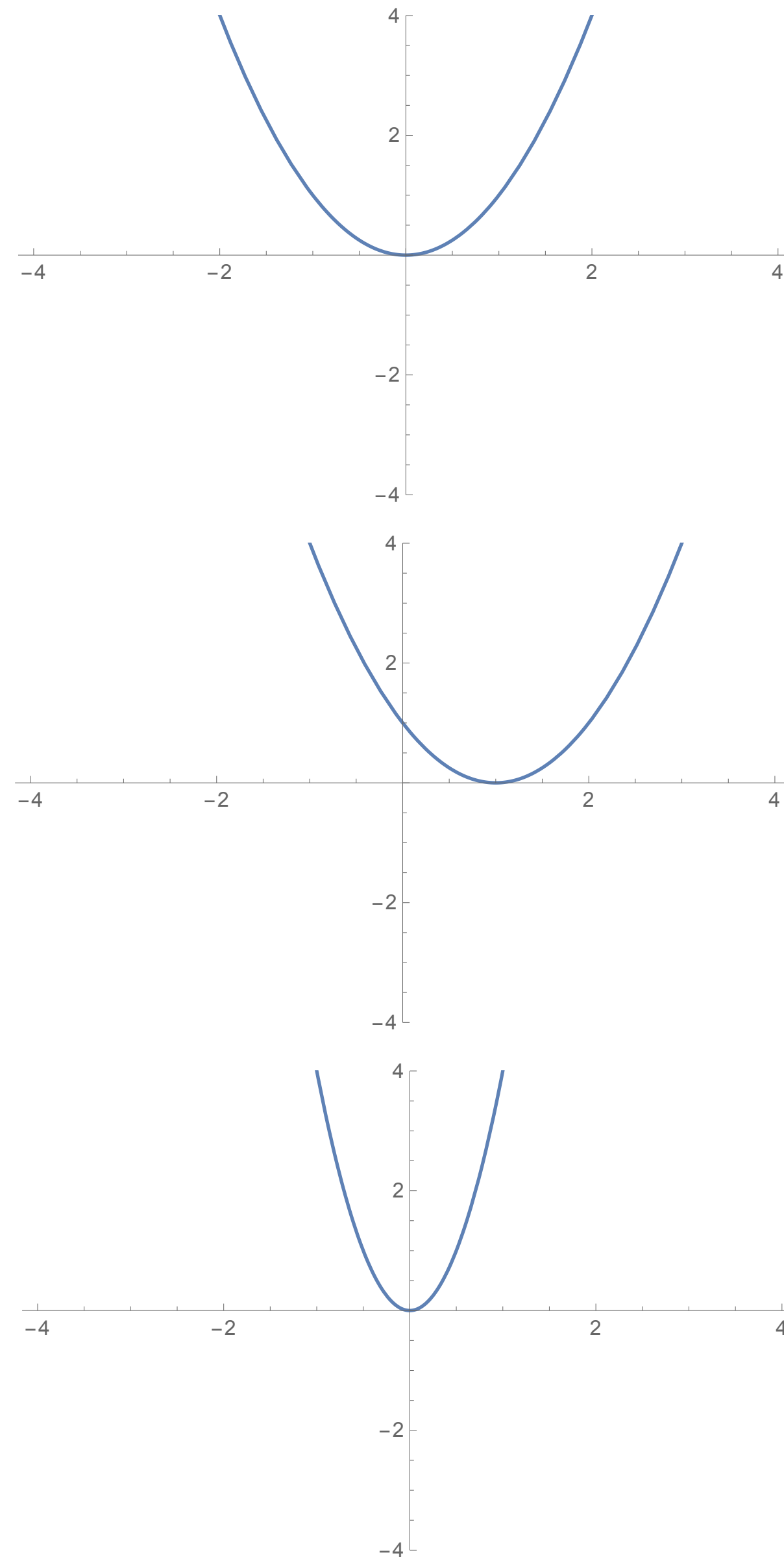
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

are matrix equivalent and row equivalent and column equivalent to

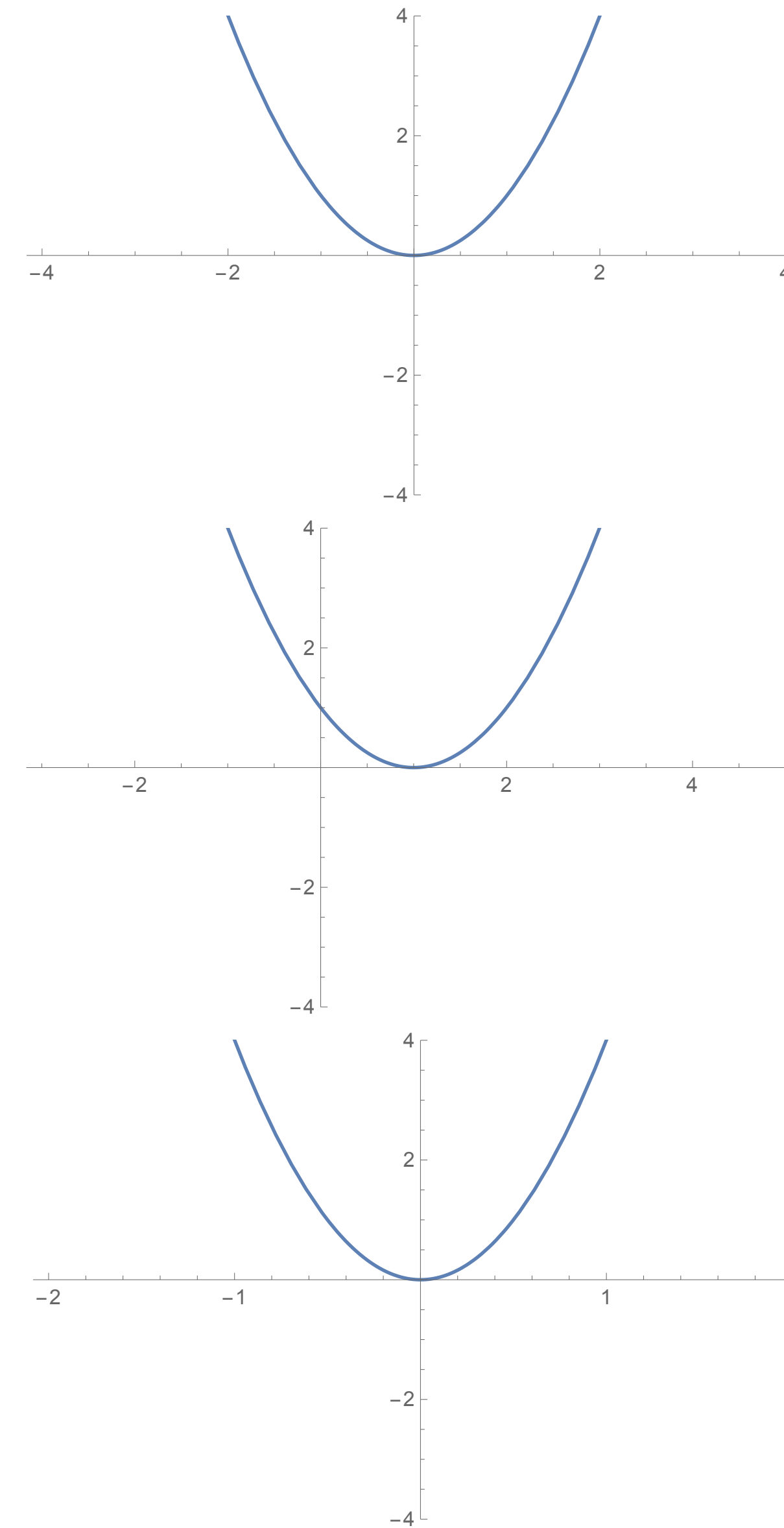
$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

But they are qualitatively different as transformations, both from each other and from the identity.

Fixed system of coordinates,
changing objects

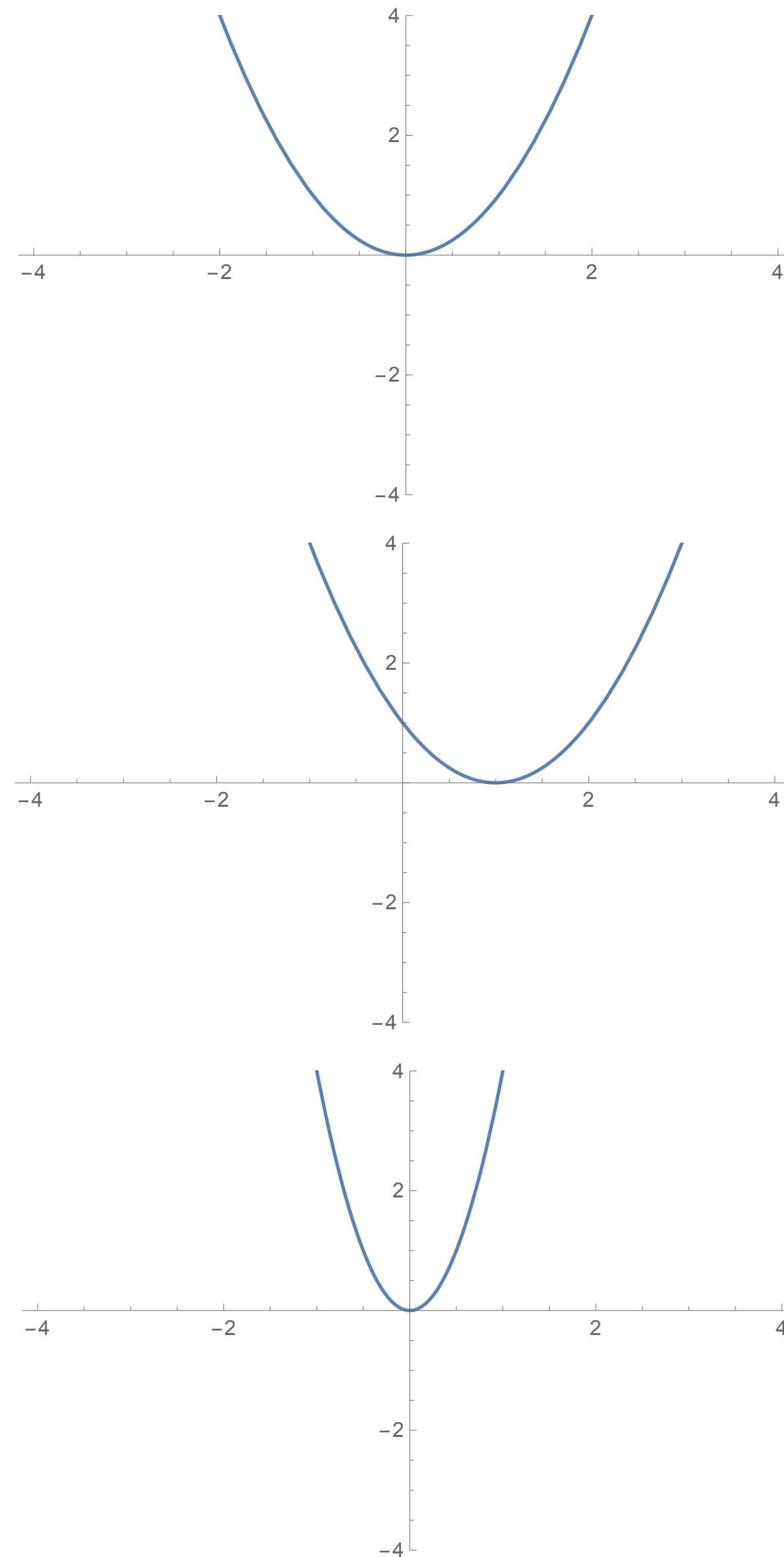


Changing system of coordinates,
fixed objects



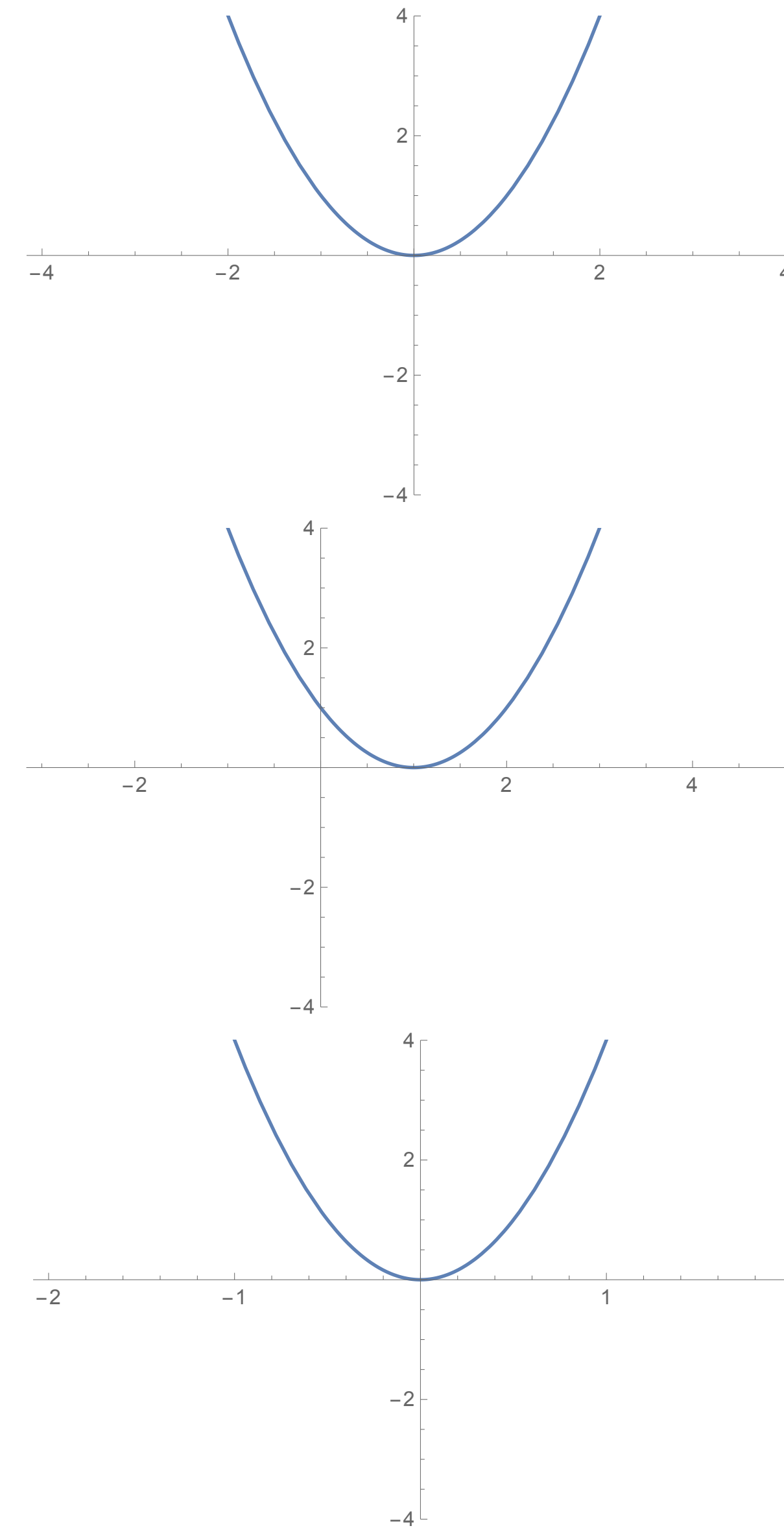
Fixed system of coordinates, changing objects

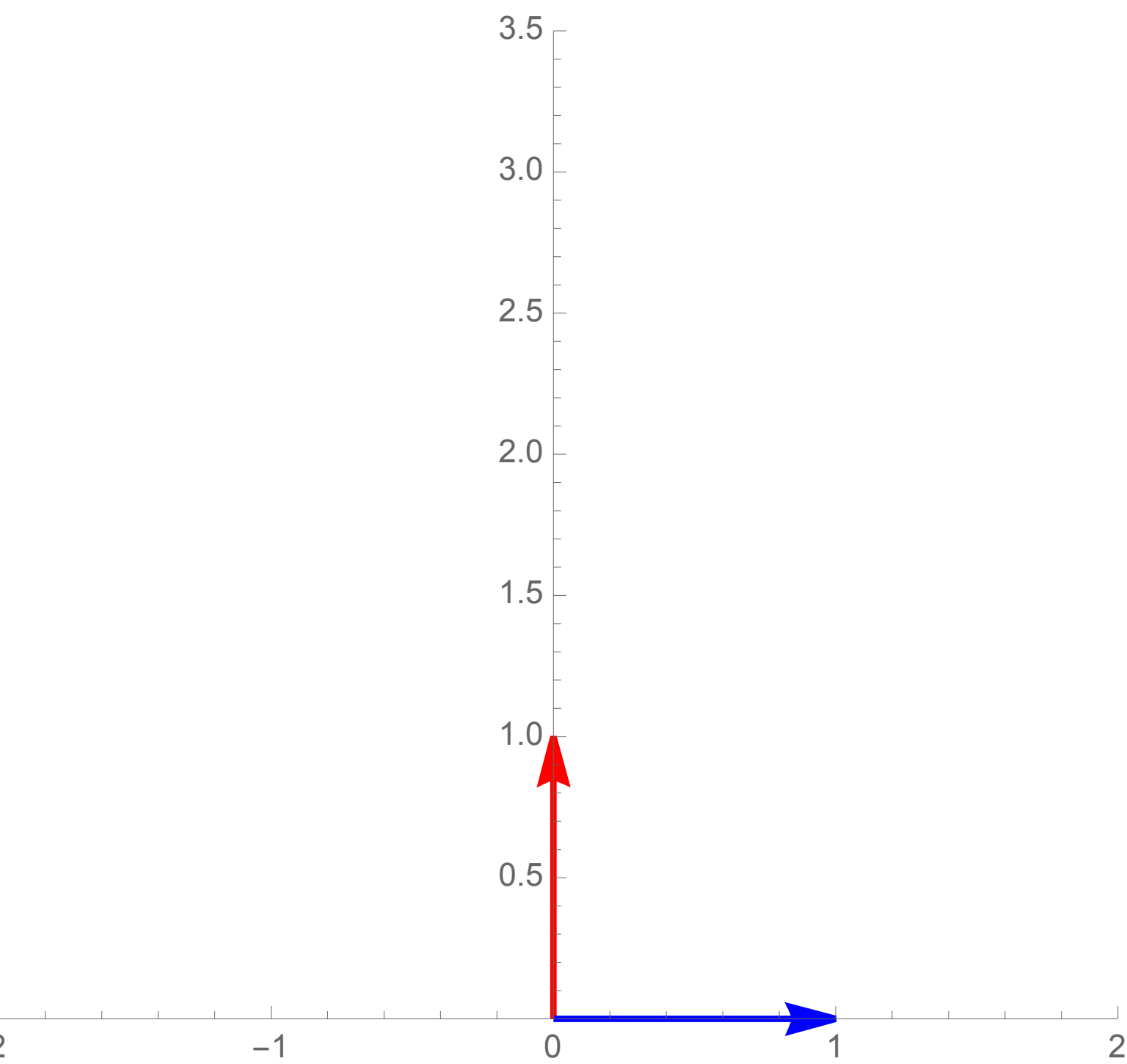
alibi



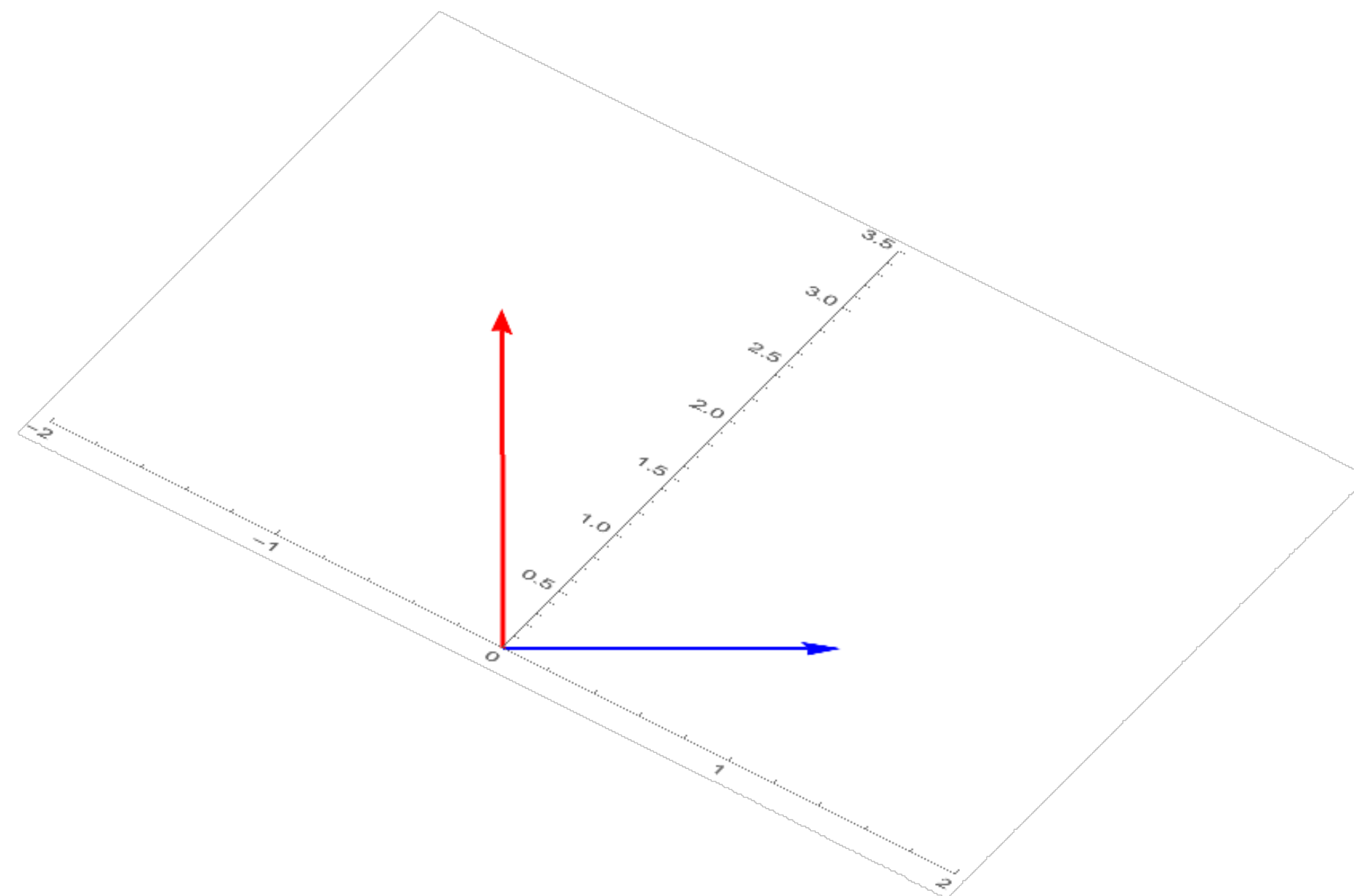
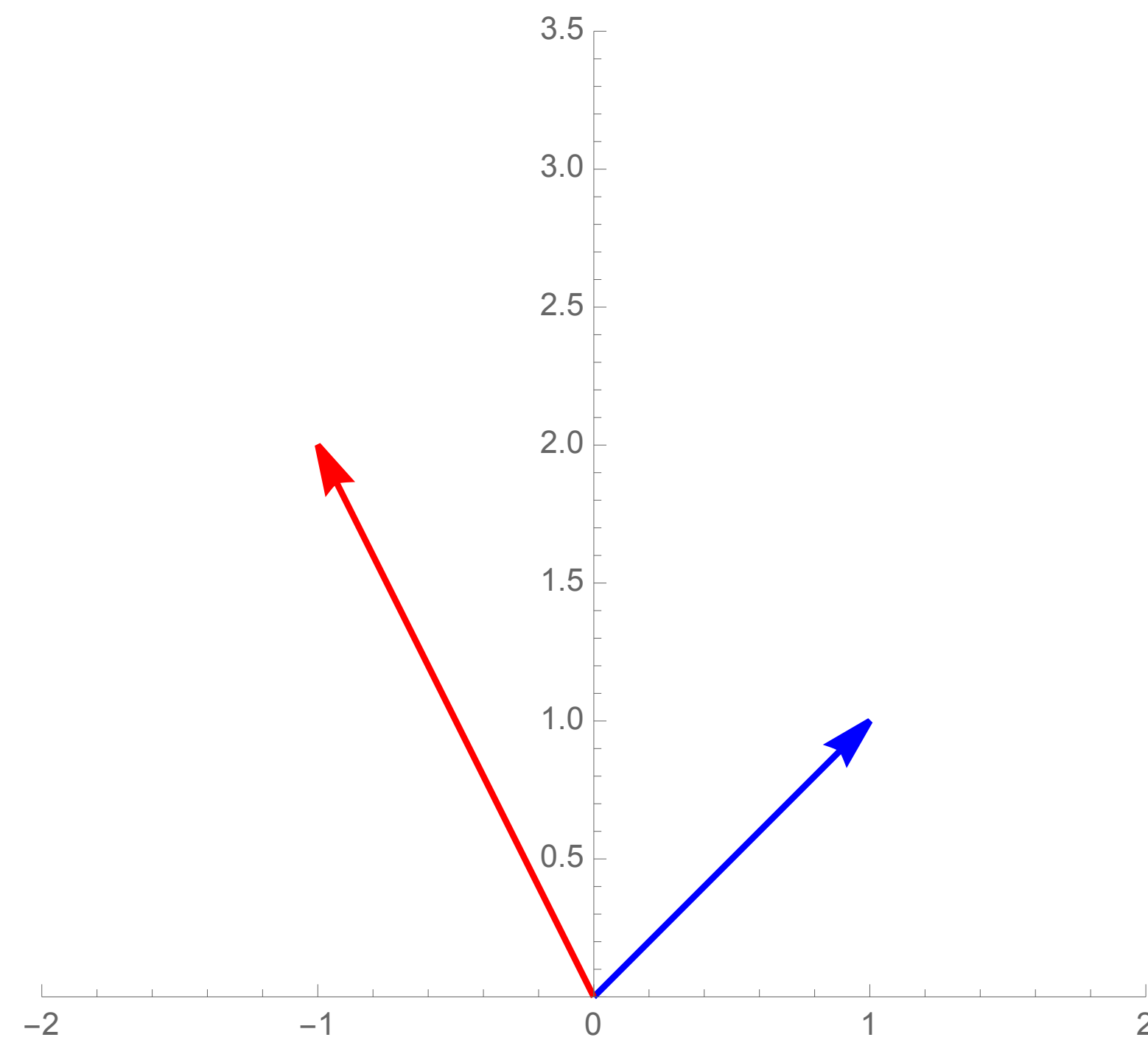
Changing system of coordinates, fixed objects

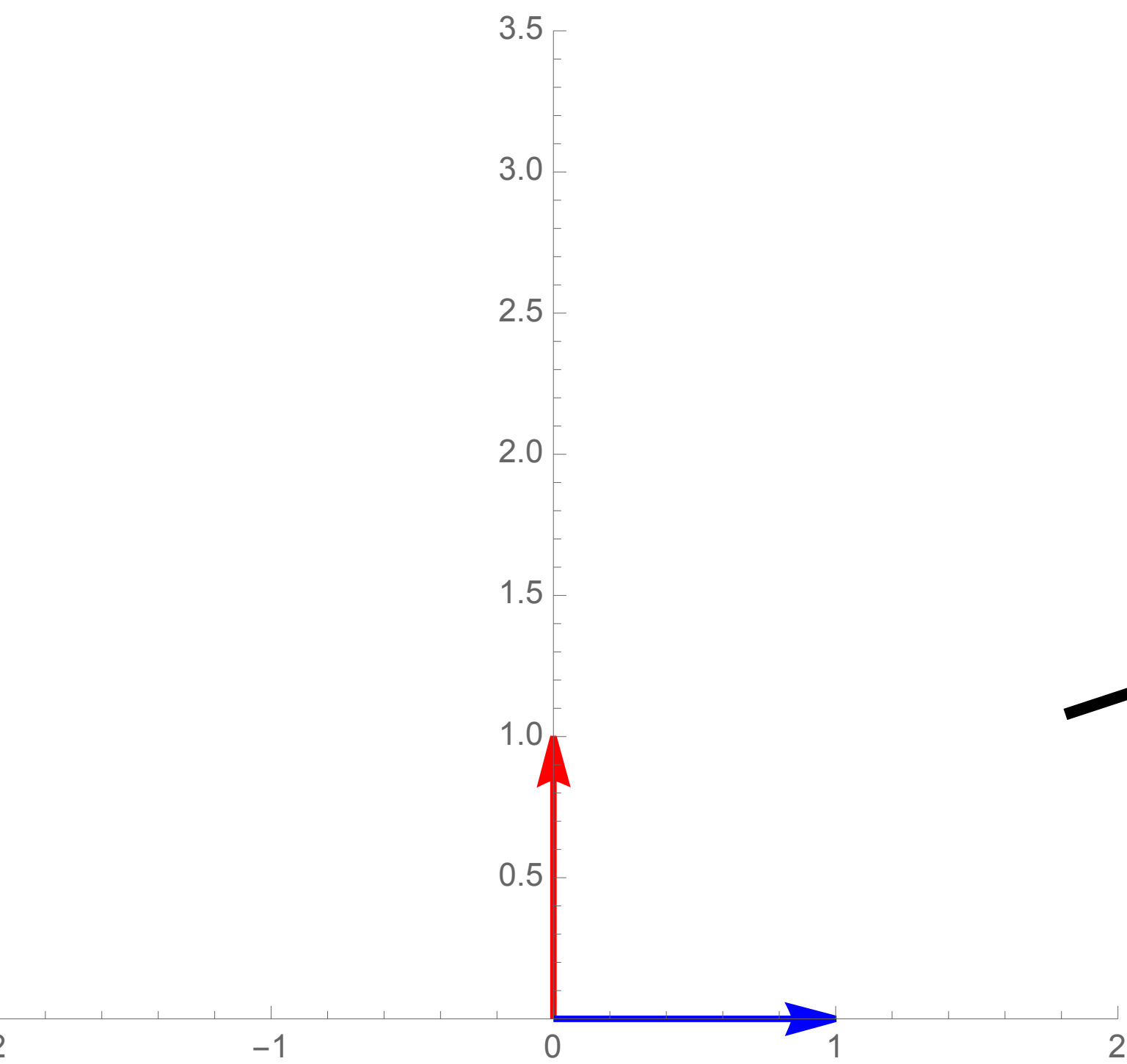
alias





$$\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

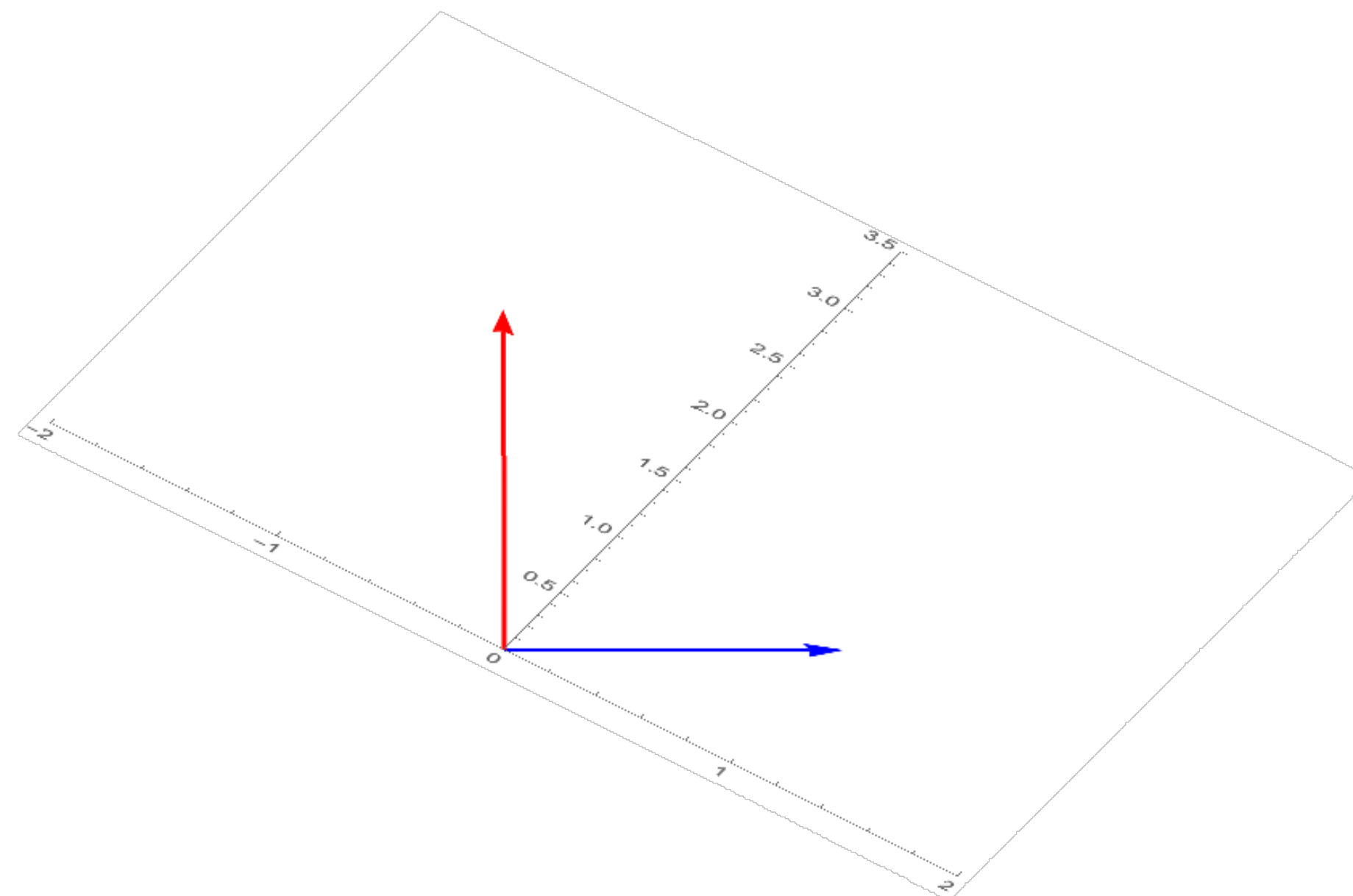
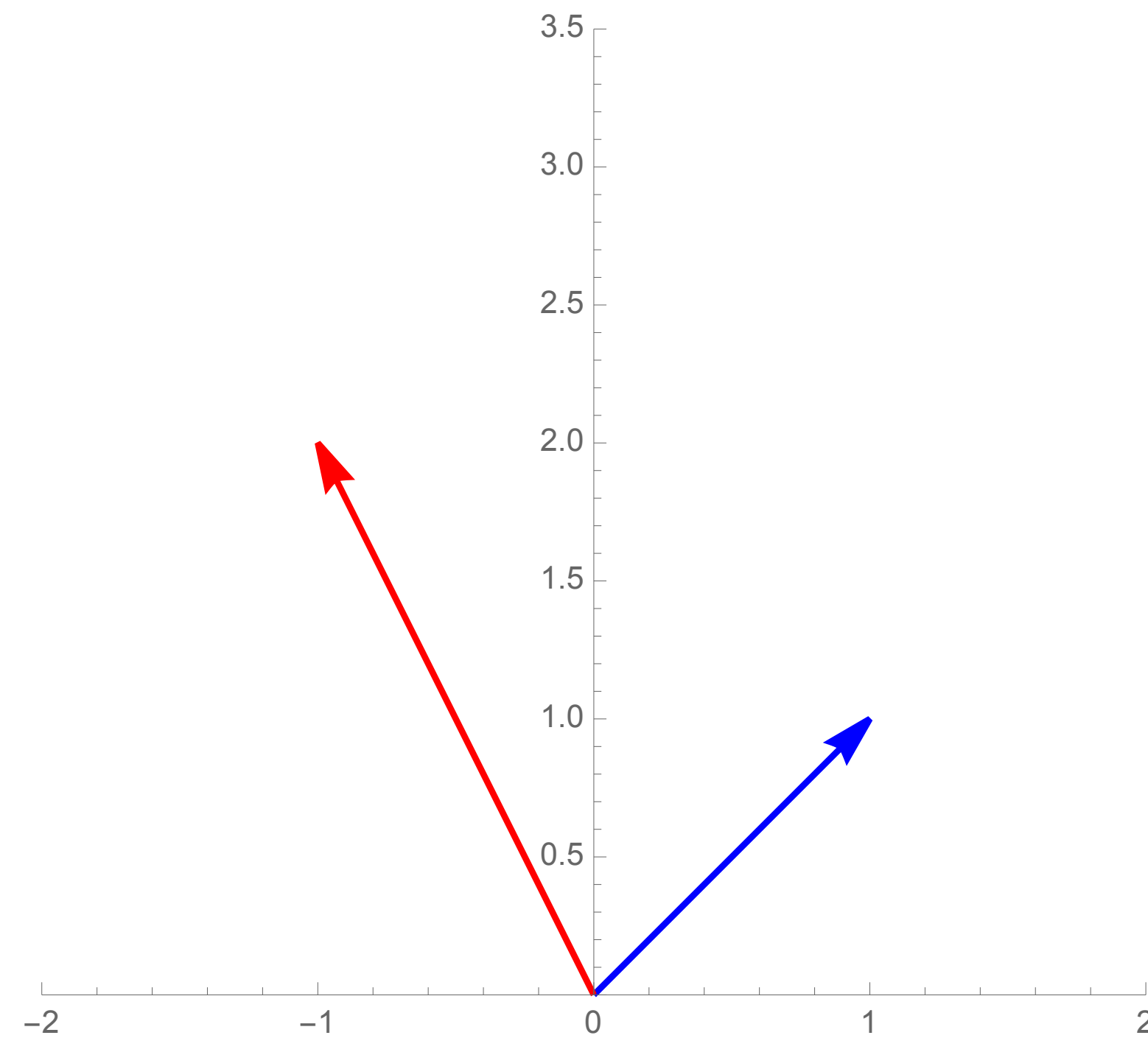




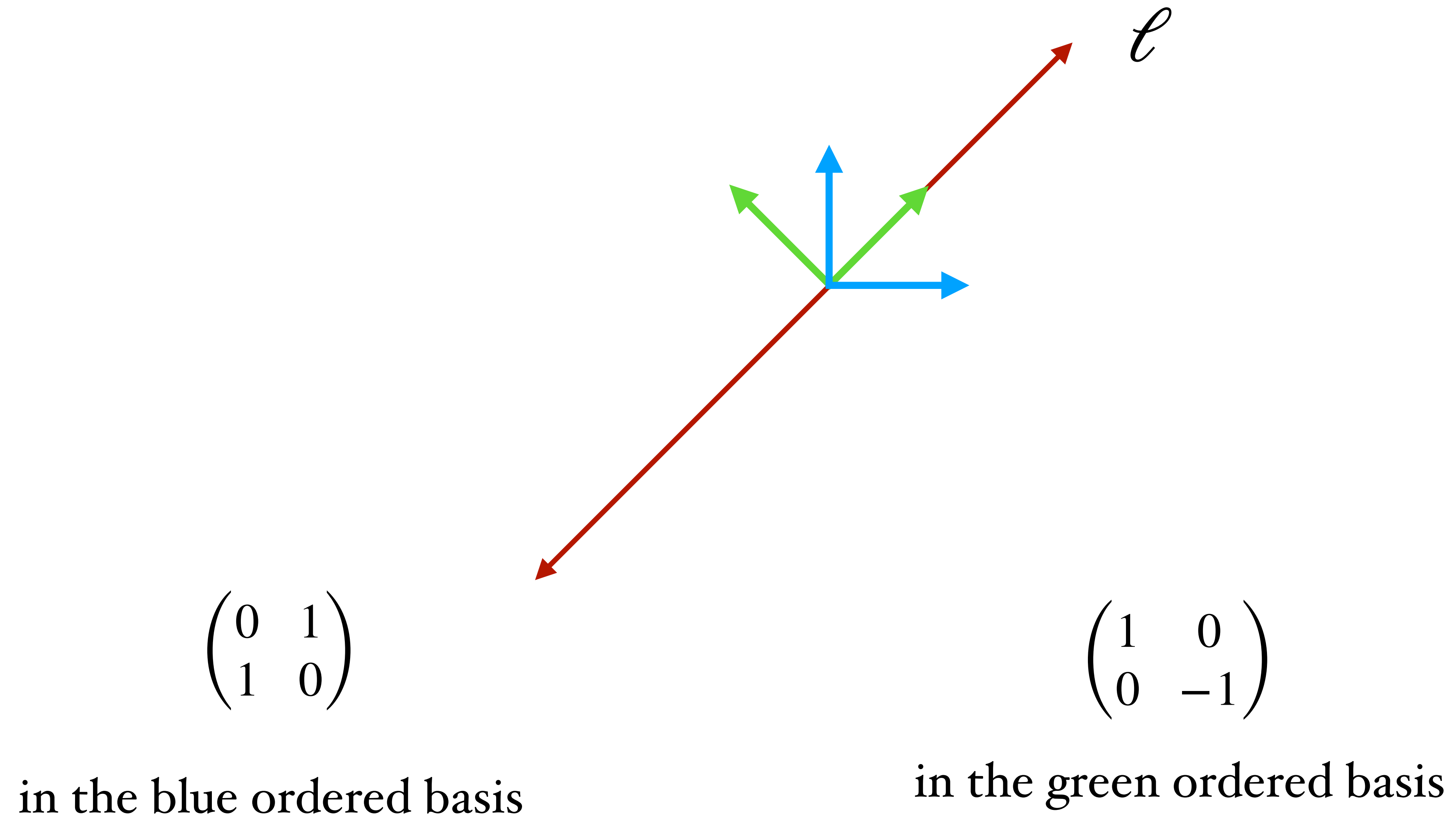
alibi

$$\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

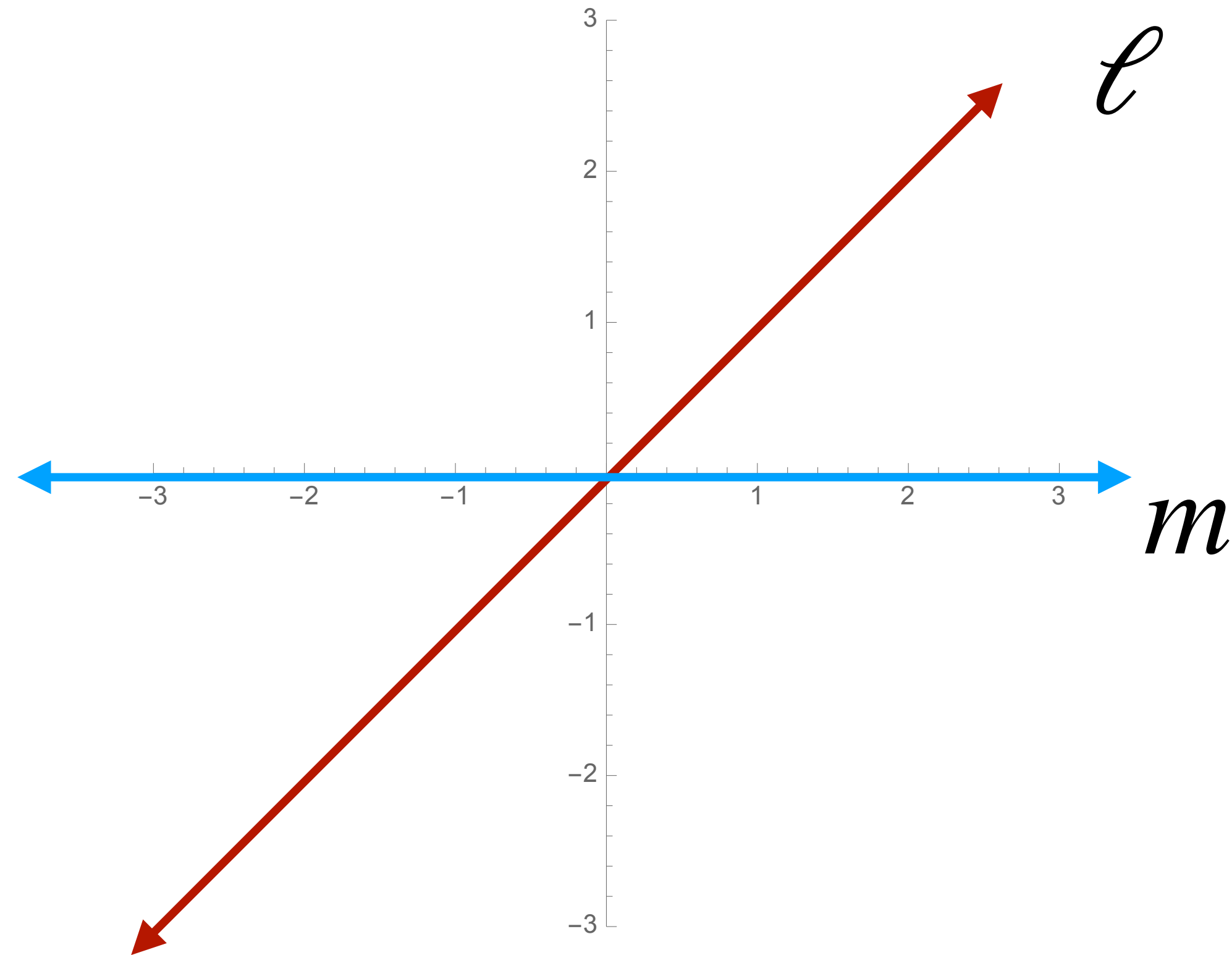
alias



Similar matrices represent the same linear transformation
in different coordinates.



Similar matrices represent different linear transformations that all have the same geometry but are located differently within the vector space, viewed with a fixed coordinate system.



$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

reflection in red line ℓ

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

reflection in blue line m

Similar matrices are related by a change of basis:

$$A \sim B \quad \text{if} \quad A = QBQ^{-1} \quad \text{for some invertible matrix } Q$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

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$$\begin{pmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Similar matrices have the same:

rank and nullity

determinant

eigenvalues

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